Jon Zorrilla Gamboa

Data Scientist & AI Engineer

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Data Scientist & AI Engineer with a Physics background and MSc in Data Science, experienced in developing, deploying, and monitoring ML/DL models using MLflow, FastAPI, and MLOps practices, with proven delivery of scalable AI solutions across scientific and industrial domains.

Experience

Ayesa — Greater Bilbao metropolitan area

Data Scientist at the Ibermática Innovation Institute (i3B). — Feb 2024 - Now

Participated in R&D projects in applied Artificial Intelligence at national and European levels, conducting research on new algorithms, data analysis, experimentation, creating predictive models, visualisations, and communicating results.

- Development of ML models to predict nickel concentration in industrial galvanizing processes, achieving an F1-score of 0.99 using industrial variables. Applied explainable AI (XAI) techniques to identify key drivers and improve model interpretability for decision-making.
- Design of a ML framework for early risk detection in construction project documentation, resulting in a scientific publication. Achieved an F1-score of 0.73 in bottleneck detection using interpretable rules, identified 6% anomalies, and segmented projects into 3 behavioural clusters. Applied clustering, ensemble anomaly detection models, and XAI to identify critical variables and risk patterns in the AEC industry.
- Development of a cloud-based tool for interactive visualisation of theoretical vs real process routes in steel production, including analysis and traceability by product category and order/position. Implemented PM4Py-based dashboards for analysing route conformity and deviations. Built MLOps pipelines with MLflow for experiment tracking and model fine-tuning in binary classification tasks. Generated synthetic data using VAEs and GANs for mixed datasets, improving model robustness.
- Development of an end-to-end hourly electricity consumption forecasting system for 80 CUPS, from data ingestion to deployment via REST API. Compared gradient boosting models, SARIMA, Prophet, and LSTM with hyper-parameter optimisation using Optuna and time series validation. Best model (CatBoost): MAE of 0.099 kWh, achieving 77% reduction vs naive baseline. Applied SHAP for interpretability and deployed using FastAPI and MLflow.
- Comparative research on classical ML vs. quantum ML (QML) for molecular classification and regression, implementing pipelines with scikit-learn, Qiskit, and MLflow. Training and evaluation of Support Vector Machines, Random Forest, and Multilayer Perceptron models against their quantum counterparts (QSVM, variational circuits, and quantum neural networks). Robustness analysis under data scarcity and noise, as well as scalability assessment of variational circuits, to determine viability thresholds and ROI for quantum computing in industrial environments.

Hubbell — Bilbao

Research Data Scientist — Oct 2023 - Feb 2024

Developed a Master's thesis titled 'Advanced Metering Infrastructure-oriented data imputation through machine learning techniques', researching and applying ML models to impute missing values in smart-meter time series from the electricity sector.

EDP Energy — Bilbao

Data Scientist — Apr 2022 - July 2022

Completed an extracurricular internship in the Business Intelligence and Big Data department of EDP Energy.

Education

Universidad Autónoma de Madrid Madrid
M.S. in Data Science 2022 - 2024

- Machine learning, deep learning, reinforcement learning, time series analysis, natural language processing, advanced statistical analysis, computer vision and advanced programming.

Universidad del País Vasco Bilbao
B.S. in Physics 2018 - 2022

Courses & Certifications

- Generative AI with Large Language Models – Coursera (AWS & DeepLearning.AI) 2025
- Machine Learning in Production – Coursera (DeepLearning.AI) 2025
- IBM Deep Learning with PyTorch, Keras and Tensorflow Professional Certificate – Coursera (IBM) 2025
- AWS Certified AI Practitioner – Amazon Web Services (AWS) 2026

Publications

J. Zorrilla, S. Seijo, U. Arenal and J. R. Mena, "AI-Driven Decision Support System for Proactive Risk Management in Construction Projects", *Intelligent Infrastructure and Construction (IIC)*, vol. 2, no. 2, p. 4, 2026. DOI: 10.3390/iic2020004

Skills

- **Programming languages:** Python, R, SQL and Fortran
- **Machine learning and Deep learning:** Numpy, Pandas, Scikit-learn, Boosting models, PyTorch, TensorFlow, PySpark, Matplotlib and Seaborn.
- **Cloud computing:** Azure and AWS.
- **Management and deployment:** Git, MLFlow and FastAPI.
- **Other:** LaTeX, OpenCV, Matlab, SAS, PowerBI and Microsoft Office.

Languages

- **Spanish:** Native.
- **English:** Cambridge Advanced Exam: C1
- **Basque:** Advanced: B2.

Projects

- **Portfolio Website** (React, Vite, Tailwind CSS).
  - Personal portfolio website for presenting professional experience and projects in Data Science and Machine Learning.
  - **Repository:** https://github.com/Jonzg/portfolio-website/
- **AWS AI Practitioner Quiz App** (FastAPI, React, SQLite, AWS).
  - Full-stack application for AWS Certified AI Practitioner exam preparation through dynamic quizzes and progress tracking.
  - **Repository:** https://github.com/Jonzg/aws-quiz/